

Topic: Basic definition, terminology and setting

Q: What is the probability? How to talk about it rigorously?

The study of **random experiments** / events.

i.e., Experiments that don't turn out the same way every time you do them.

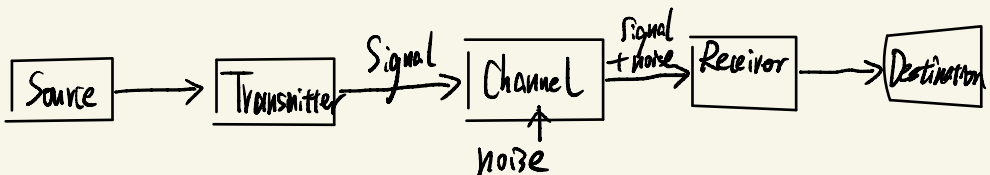
Opposite to the **deterministic procedures** / ^{experiments} that always give the same results, e.g., math / physics laws, $E = IR$
Ohm's law.

Random experiments are everywhere / Applications of probability:

- Games, Gambling
- Financial markets (stock market, insurance company)
- Simulations of large systems (transportation system, power system)
- Signal and image processing
- Communication and information theory
- Machine learning / Deep learning

ECSE
Courses-related

□ A concrete EE example of random experiments



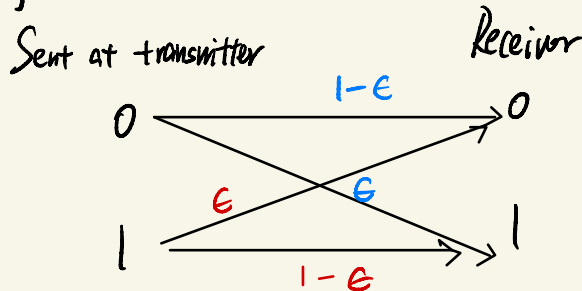
This model holds for many EE/CSE systems :

- cell phones (voice, text, image)
- Wireless network (wifi, bluetooth)
- TV (cable)
- Satellite systems

In those systems, messages at receiver come with noise, where the noise come from

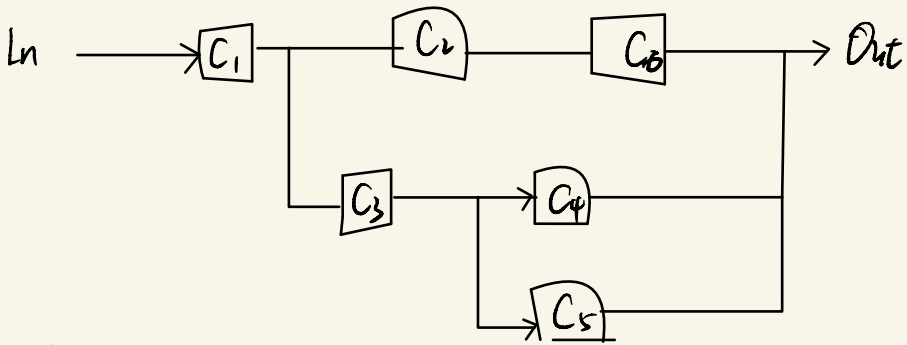
- Atmosphere interference (weather, fading)
- Acoustic noise
- Quantization noise
- Signal crosstalk

□ A good model for many digital communication system is the binary channel :



- What is $P(0 \text{ sent} / 0 \text{ received})$?
- How can we modify the coder/decoder to decrease the error probability?
- How can we compare the expected performance of different communication strategies?

□ A complex system example (computer, circuit system)

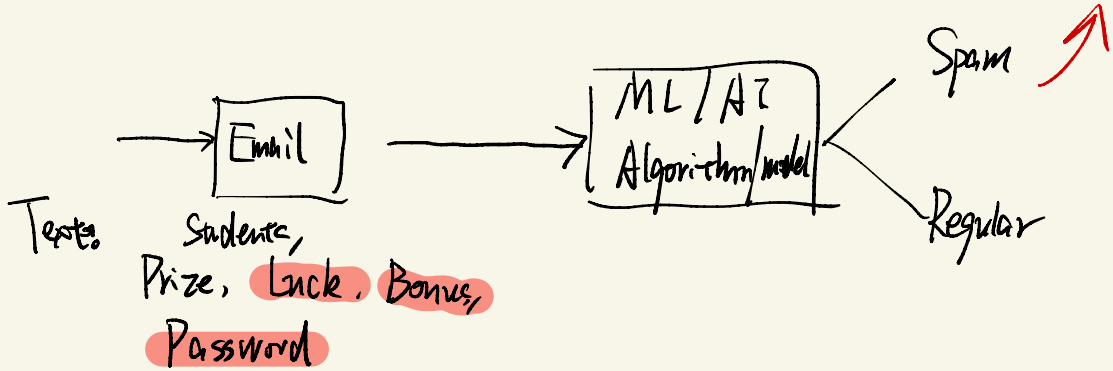


Each component has a probability of failure within T hours.

- What is the expected time to failure of the overall system?
- What is the probability that 3 out of the 6 components fail within the same interval?
- What is the probability that the system can last 6 months without failure?

□ A machine learning example (Natural language processing)

Spam email classification



- What is the probability of a spam email if it contains the word "Bonus", "Password", "Cash"?

□ Many times, we can simply answer those complicated questions related to probability using very basic models:

- ① Flip a coin $\begin{matrix} \swarrow \text{Head} \\ \searrow \text{Tail} \end{matrix}$
- ② Roll a Die $\begin{matrix} \swarrow \frac{1}{6} \\ \searrow \frac{2}{6} \\ \quad \frac{3}{6} \\ \quad \frac{4}{6} \\ \quad \frac{5}{6} \\ \quad \frac{6}{6} \end{matrix}$
- ③ Pick a ball out of a blind bag

